

CampaignIQ

Marketing Campaign Optimizer

Final Project Report
End-to-End Marketing Experimentation & A/B Testing Analytics Platform

Technology: Python • SQL • SQLite • Statistics • Power Analysis • Streamlit • Power BI •
DAX

Author
Thanuj Vummidi

Project scope and terminology follow the CampaignIQ project plan and completed implementation.

1. Executive Summary

CampaignIQ is an end-to-end marketing experimentation platform that evaluates A/B tests using conversion or retention metrics, statistical significance, confidence intervals, power analysis, sample-size requirements, and an automated verdict engine. The completed solution connects data ingestion, exploratory analysis, a SQLite analytical layer, reusable statistical modules, a Streamlit application, and a three-page Power BI dashboard.

The project uses the Marketing A/B Testing dataset for the primary marketing analysis and the Cookie Cats experiment as an independent validation case. It also generates 40 simulated campaigns across 10 marketing channels, with 5,000 control and 5,000 treatment users per campaign.

Project Metric	Completed Result
Marketing A/B Testing users	588,101
Cookie Cats users	90,189
Simulated campaigns	40
Simulated experiment records	400,000
Experiment results stored	40
Statistically significant simulated results	10
Underpowered simulated experiments	40
Production statistical validation	PASS

2. Business Problem

Marketing teams run experiments to determine whether a new campaign, advertisement, message, or strategy improves customer behavior. A treatment can show a higher observed conversion rate while the evidence remains insufficient to distinguish the effect from sampling variation.

CampaignIQ therefore addresses a broader set of questions: Did treatment outperform control? How large was the observed effect? Is the difference statistically significant? What is the confidence interval? Was the experiment adequately powered? What sample size would be required to detect a predefined minimum detectable effect?

3. Project Objectives

- Build a validated data-ingestion and preprocessing workflow.
- Perform exploratory data analysis on marketing and retention experiments.
- Create reusable statistical testing, power-analysis, and verdict modules.
- Create a SQLite analytical source of truth for experiment data and results.
- Generate a portfolio of 40 simulated marketing campaigns across multiple channels.
- Provide an interactive Streamlit application for existing and uploaded experiments.
- Build a three-page Power BI dashboard for executive, campaign, and channel analysis.
- Independently validate the production statistical engine against the Cookie Cats experiment.

4. Data Sources

4.1 Marketing A/B Testing Dataset

The processed Marketing A/B Testing dataset contains 588,101 user-level records. The relevant fields are user_id, test_group, converted, total_ads, most_ads_day, and most_ads_hour. The raw identifier column was removed during processing and source column names were standardized.

4.2 Cookie Cats Dataset

The processed Cookie Cats dataset contains 90,189 user-level records and is used for independent validation. The fields include userid, version, sum_gamerounds, retention_1, and retention_7.

4.3 Simulated Campaign Portfolio

CampaignIQ generates 40 campaigns across Email, Social Media, Search, Display, Push Notification, SMS, In-App, Website, Referral, and Loyalty. Each campaign contains 5,000 control and 5,000 treatment records, producing 400,000 simulated experiment records.

5. Data Quality & Validation

Validation	Marketing A/B	Cookie Cats
Missing values	0	0
Duplicate full rows	0	0
Unique user IDs	Equal to row count	Equal to row count
Core value validation	Passed	Passed

The ingestion notebook validates schemas, missing values, duplicate records, identifiers, and domain values before writing the processed datasets.

6. Statistical Methodology

6.1 Conversion / Retention Rate

$$\text{Rate} = \text{Successes} / \text{Total Users}$$

6.2 Absolute Difference

$$\text{Absolute Difference} = \text{Treatment Rate} - \text{Control Rate}$$

6.3 Relative Lift

$$\text{Relative Lift} = (\text{Treatment Rate} - \text{Control Rate}) / \text{Control Rate}$$

6.4 Hypothesis Test

For binary conversion and retention metrics, the production engine uses a two-proportion z-test. The default significance level is $\alpha = 0.05$. The null hypothesis states that the treatment and control rates are equal; the two-sided alternative states that they are different.

6.5 Confidence Interval

The production significance module reports a confidence interval for the treatment-control difference. This complements the p-value by showing the estimated effect and its uncertainty.

6.6 Power and Sample Size

The power calculator uses a baseline rate, relative minimum detectable effect, alpha, and target power. The configured project example uses alpha = 0.05, power = 0.80, and a 20% relative MDE.

7. Marketing A/B Testing Analysis

Metric	Result
Control conversion rate	1.7854%
Treatment conversion rate	2.5547%
Absolute difference	+0.7692 percentage points
Relative lift	+43.09%
Z-statistic	7.3701
P-value	1.705281e-13
95% CI for difference	[+0.5951 pp, +0.9434 pp]

The observed treatment conversion rate exceeded the control rate. Under the configured 5% significance level, the observed difference was statistically significant.

7.1 Exposure Analysis

The exploratory analysis also examined total advertisement exposure. The distribution is strongly right-skewed: the mean total_ads value is 24.82, the median is 13, and the maximum is 2,065. The analysis retains the distribution rather than blindly deleting high-exposure observations.

8. Cookie Cats Validation Results

Metric	Day-1 Retention	Day-7 Retention
Control rate	44.8188%	19.0201%
Treatment rate	44.2283%	18.2000%
P-value	0.0744096553	0.0015542500
95% CI	[-1.2392%, +0.0582%]	[-1.3282%, -0.3121%]
Production validation	PASS	PASS

The independent validation reproduced the notebook calculations in the production statistical engine. Day-1 retention did not meet the configured 5% significance threshold, while Day-7 retention did.

9. Production Statistical Engine

The production implementation is separated into reusable modules. `significance_test.py` performs the two-proportion test and confidence interval calculation. `power_calculator.py` estimates required sample size. `verdict_engine.py` combines statistical evidence and sample adequacy into an experiment interpretation.

```
scripts/
■■■ significance_test.py
■■■ power_calculator.py
■■■ verdict_engine.py
■■■ campaign_simulator.py
■■■ load_database.py
■■■ generate_experiment_results.py
```

The validation notebook recalculates Day-1 and Day-7 Cookie Cats results from scratch and compares control rate, treatment rate, p-value, and confidence-interval bounds with the production engine. All checks passed.

10. Power Analysis Findings

Each simulated campaign contains 5,000 users per group. Under the configured 20% relative MDE, $\alpha = 0.05$, and 80% power requirement, all 40 simulated campaigns were below the calculated required sample size. Required sample sizes varied substantially by baseline rate.

Portfolio measure	Value
Actual sample per group	5,000
Configured relative MDE	20%
Alpha	0.05
Target power	0.80
Underpowered campaigns	40 / 40
Example highest required per-group sample	46,352
Example lower required per-group sample	8,040

This illustrates that statistical significance and power are distinct concepts. A result can reach statistical significance while still failing a predefined sample-size requirement for a target MDE.

11. Automated Verdict Engine

The verdict engine uses p-value, alpha, treatment and control rates, confidence-interval bounds, actual sample size, and required sample size. It distinguishes statistically significant positive and negative results from non-significant outcomes and communicates sample inadequacy separately.

The implementation explicitly avoids treating a non-significant result as proof that treatment and control are identical.

12. SQL & Data Architecture

SQLite serves as the analytical source of truth. The database contains marketing_ab, cookie_cats, campaigns, and experiment_results tables. The experiment_results table stores experiment-level statistical outputs used by downstream applications and reporting.

```
db/marketing_campaign_optimizer.db
```

```
Tables:
marketing_ab
cookie_cats
campaigns
experiment_results
```

The campaigns table stores campaign metadata together with user-level experiment observations. The experiment_results table stores campaign-level statistical results, including rates, lift, p-values, confidence intervals, sample sizes, significance, and decisions.

13. Streamlit Application

The Streamlit application provides an interactive interface for selecting an existing campaign or uploading a custom CSV experiment. It displays control and treatment conversion, relative lift, p-value, confidence interval, significance status, sample adequacy, interpretation, and experiment planning information.

```
streamlit run app\app.py
```

The CSV upload workflow expects user-level fields user_id, test_group, and converted.

14. Power BI Dashboard

The completed Power BI dashboard contains three pages designed around different analytical levels.

Page	Purpose	Main Content
Experiment Overview	Portfolio view	KPIs, campaign lift, significance, sample adequacy, channel view, evidence

Page	Purpose	Main Content
Campaign Performance	Campaign-level analysis	Campaign slicer, conversion metrics, lift, p-value, confidence interval, evidence
Channel Performance	Channel-level analysis	Relative lift by channel, campaigns by channel, channel summary, slicer

The dashboard uses the Power BI-ready analytical data package derived from the SQLite database. Interactive slicers are used to filter campaign and channel views.

15. Key Findings

- **Marketing A/B Test:** treatment conversion was 2.5547% versus 1.7854% for control, corresponding to +43.09% observed relative lift.
- **Marketing statistical evidence:** $p = 1.705281e-13$ and the 95% CI for the difference was [+0.5951 pp, +0.9434 pp].
- **Cookie Cats Day-1:** $p = 0.0744096553$, with a confidence interval spanning zero.
- **Cookie Cats Day-7:** $p = 0.0015542500$, with the confidence interval entirely below zero for the treatment-control difference.
- **Simulated portfolio:** 40 campaigns and 400,000 records were generated, with 40 experiment-result records stored.
- **Sample adequacy:** all 40 simulated experiments were underpowered for the configured 20% relative MDE and 80% target power.
- **Validation:** notebook and production statistical results matched for both Cookie Cats metrics.

16. Limitations & Considerations

- The simulated campaign portfolio is intended for demonstrating the analytics workflow and is not real campaign performance data.
- Power calculations depend on the selected baseline rate, MDE, alpha, and target power.
- Statistical significance does not by itself establish practical or business significance.
- A non-significant result should not be interpreted as proof of no difference.
- The current project focuses on two-group proportion testing; advanced methods such as sequential testing and Bayesian experimentation are future extensions.
- Multiple-testing correction should be applied when a workflow makes many simultaneous statistical claims; the project plan identifies Bonferroni correction as an appropriate option when required.

17. Final Project Structure

```
marketing_campaign_optimizer/
├── data/
│   ├── raw/
│   └── processed/
├── db/
├── notebooks/
│   ├── 01_data_ingestion.ipynb
│   ├── 02_eda_analysis.ipynb
│   └── 03_validation_report.ipynb
├── scripts/
│   ├── campaign_simulator.py
│   ├── significance_test.py
│   ├── power_calculator.py
│   ├── verdict_engine.py
│   └── load_database.py
```

```

■ ■■■ generate_experiment_results.py
■ ■■■ sql/
■ ■■■ schema.sql
■ ■■■ queries.sql
■ ■■■ app/
■ ■■■ app.py
■ ■■■ dashboard/
■ ■■■ campaign_optimizer_dashboard.pbix
■ ■■■ reports/
■ ■■■ Final_Report.pdf
■ ■■■ Presentation.pptx
■ ■■■ requirements.txt
■ ■■■ README.md

```

18. Technology Stack

Area	Technology
Programming	Python
Data analysis	Pandas, NumPy
Statistics	SciPy, Statsmodels
Visualization	Matplotlib, Seaborn, Plotly
Database	SQLite
Analytics	SQL
Application	Streamlit
Business Intelligence	Power BI, DAX
Notebook	Jupyter Notebook
Development	VS Code, PowerShell
Version control	Git / GitHub

19. Reproduction Guide — Windows 11

```

python -m venv .env
.\.env\Scripts\Activate.ps1
pip install -r requirements.txt
jupyter notebook

```

Run the notebooks in sequence: 01_data_ingestion.ipynb, 02_eda_analysis.ipynb, and 03_validation_report.ipynb.

```

python scripts\campaign_simulator.py
python scripts\load_database.py
python scripts\generate_experiment_results.py
streamlit run app\app.py

```

20. Future Enhancements

- Bayesian A/B testing
- Sequential testing
- CUPED variance reduction
- Multi-armed bandits
- Automated experiment monitoring
- Revenue-based experimentation metrics

- Customer segmentation
- Experiment history tracking
- Cloud database integration and deployment
- Automated data refresh and reporting
- Expanded multiple-testing correction workflows
- Experiment anomaly detection

21. Conclusion

CampaignIQ delivers the planned end-to-end marketing experimentation workflow: raw data is validated and transformed, exploratory analysis is performed, experiment data is stored in SQLite, statistical tests and power calculations are executed through reusable production modules, results are surfaced through Streamlit, and the analytical portfolio is presented through Power BI.

The completed validation demonstrates that the production statistical engine agrees with the independent notebook calculations for the Cookie Cats Day-1 and Day-7 retention experiments. The project therefore provides a reproducible foundation for evaluating marketing experiments while keeping statistical evidence, confidence intervals, and sample adequacy visible alongside observed performance.

22. Author

Thanuj Vummidi

Aspiring Data Analyst / Data Engineer

Areas of interest: Python, SQL, Data Analytics, Data Engineering, A/B Testing, Statistical Analysis, Power BI, Microsoft Fabric, and Cloud Data Engineering.